

Quasinormal modes analysis of a massive scalar field in a Schwarzschild background via the Spectral Method

Davide Batic,^{1,*} Anna Chrysostomou,^{2,†} Alan S. Cornell,^{3,‡} and Denys Dutykh^{1,§}

¹ *Mathematics Department, Khalifa University of Science and Technology,
PO Box 127788, Abu Dhabi, United Arab Emirates*

² *Laboratoire de Physique Théorique et Hautes Énergies - LPTHE,
Sorbonne Université, CNRS, 4 Place Jussieu, 75005 Paris, France*

³ *Department of Physics, University of Johannesburg,
PO Box 524, Auckland Park 2006, South Africa*

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We study the complex quasinormal frequencies ω of a massive scalar field of mass μ , which are known to transition from propagative quasinormal modes to evanescent quasinormal modes when $\Re[\omega^2] \lesssim \mu^2$. In this regime, well-established semi-analytical and numerical methods (e.g. WKB and continued-fraction approaches) break down due to the modified asymptotic behaviour induced by the mass term. We develop an extended spectral method in which the spectral discretisation yields a seventh-degree polynomial eigenvalue problem in ω , reflecting the nontrivial analytic structure of the massive dispersion relation. By implementing boundary conditions consistent with this structure via a modified quasinormal mode ansatz, we obtain stable and accurate spectra beyond the conventional $\ell < n$ and $\Re[\omega^2] < \mu^2$ regimes. We further identify branches of purely imaginary modes that emerge with increasing spectral resolution (i.e. larger Chebyshev basis size), which indicate the presence of additional mode families missed by extant methods.

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I. INTRODUCTION

As a perturbed black hole of mass M relaxes towards equilibrium, its intermediate “ringdown” phase will be dominated by characteristic quasinormal modes (QNMs). As demonstrated in the seminal works of Vishveshwara [1, 2], Press [3], and Leaver [4], the behaviour of these modes can be described as an intrinsically dissipative but well-defined boundary value problem, with purely ingoing modes at the black hole event horizon and purely outgoing modes at spatial infinity. The corresponding eigenvalues, *viz.* the discrete spectrum of quasinormal frequencies (QNFs),

$$\Omega = \Omega_R + i\Omega_I, \quad \Omega = M\omega, \quad (1)$$

are comprised of a real (Ω_R) component representing the physical oscillation frequency and an imaginary component ($\Omega_I < 0$) corresponding to the exponential decay rate. For static, spherically-symmetric black hole spacetimes, QNFs for fields of spin s depend on a multipolar value ℓ ; for each ℓ , there is a countable infinity of QNFs labelled by the overtone number n . The eikonal (geometric optics) limit corresponds to $\ell \rightarrow \infty$, while the highly-damped regime is defined by the large overtone limit $n \rightarrow \infty$.

The QNMs of massless scalar fields propagating on a Schwarzschild background have been explored extensively in the literature (see Refs [5, 6], and the references therein), and have come to be well understood. However, introducing a nonzero field mass $\mu = mM$ modifies the QNM eigenvalue problem by altering the asymptotic structure of the radial equation, thereby changing the admissible boundary conditions¹ and the physical interpretation of the resulting spectrum, necessitating a more careful treatment [7–11]. Specifically, the effective QNM potential acquires an additional mass term, elevating its asymptotic value from zero to μ^2 . As the field mass increases, the barrier height relative to this asymptotic value is reduced, and the potential progressively approaches a step-like profile. Once the asymptotic value exceeds the local maximum, the trapping region disappears and QNM behaviour can no longer

*Electronic address: davide.batic@ku.ac.ae

†Electronic address: chrysostomou@jussieu.lpthe.fr

‡Electronic address: acornell@uj.ac.za

§Electronic address: denys.dutykh@ku.ac.ae

¹ The purely outgoing behaviour that defines massive scalar QNMs can only be imposed within a restricted region of parameter space. Beyond this, we must instead impose a decaying condition at infinity, which yields a spectrum of quasibound states localised near the black hole [7]. We shall discuss this in further detail in the next section.

be supported. The spectrum is then dominated by quasibound states and evanescent modes, with the transition marked by the emergence of quaresonances – arbitrarily long-lived modes whose damping rate tends to zero [8, 9]. These quaresonances provide a continuous connection between dissipative trapped modes and effectively bound configurations. Note that we distinguish between “propagative modes”, for which $\Re[\omega^2] > \mu^2$, and “evanescent modes”, for which $\Re[\omega^2] < \mu^2$ [12].

The introduction of a mass scale also imprints characteristic signatures on the late-time dynamics. Unlike massless perturbations, which decay via inverse power-law tails governed by backscattering from asymptotic curvature, massive scalar fields exhibit oscillatory late-time tails with frequencies set by the field mass [7, 8]. These tails dominate the intermediate and asymptotic evolution and are associated with the branch cut structure of the Green’s function in the complex frequency plane. This same analytic structure underlies the multi-valued behaviour of the massive dispersion relation, which must be treated consistently at the level of the QNM eigenvalue problem. Physically, they arise from the subluminal propagation of massive excitations, which can remain localised far from the black hole and produce slowly decaying signals. The same mass term also generates an outer potential well that supports quasibound states. As such, the structure of the massive QNM eigenvalue problem encodes information beyond QNFs, providing insight also into late-time tails and long-lived modes.

Semi-analytic and numerical approaches such as (higher-order) WKB techniques and continued-fraction methods (CFMs) have proven highly effective for massless perturbations, but their applicability is reduced in the presence of a nonzero field mass. As carefully reviewed in Ref. [13], WKB schemes rely on a well-defined potential barrier and lose accuracy as the barrier is suppressed and the potential approaches a step-like profile, while CFM implementations become numerically unstable when the asymptotic boundary conditions deviate from the standard purely outgoing form. Consequently, important features of the massive QNM spectrum remain inaccessible within these frameworks.

Recently, (pseudo)spectral methods have gained traction for their exponential convergence and ability to resolve modes within the parameter space where traditional methods fail. The application of these methods begins with the discretisation of the perturbation equations into a global spectral basis, where derivatives and operators are represented as matrices acting on the spectral coefficients; the QNM eigenvalue problem is thereby transformed into a matrix eigenvalue problem that is quadratic in the frequency for a massless scalar field in a Schwarzschild background. This formulation enables highly accurate computations with a controllable number N of basis functions, which allows for the systematic extraction of both fundamental modes and higher overtones, as well as a robust treatment of nontrivial boundary conditions and asymptotic structures. While computationally more demanding than standard methods, particularly as we increase N , spectral approaches remain stable even in regimes exhibiting overdamped branches or nontrivial asymptotic structure (see, for example, Refs [14–17]).

Several variants of (pseudo)spectral methods have been proposed for QNM computations, including Chebyshev and Bernstein expansions, collocation schemes, and global polynomial discretisations (see, for example). In this work, we adopt the Chebyshev-based approach developed in Refs [18–23], which provides a flexible and robust framework for solving the massless problem. The introduction of a nonzero scalar field mass μ , however, modifies both the analytic and numerical structure of the eigenvalue problem. In the standard formulation, factoring out the asymptotic behaviour and discretising the radial equation yields a quadratic matrix eigenvalue problem of the form

$$P_2(\Omega) \mathbf{u} = (\Omega^2 M_2 + \Omega M_1 + M_0) \mathbf{u} = 0. \quad (2)$$

For the massive field, the asymptotic wavenumber becomes

$$k(\Omega) = \sqrt{\Omega^2 - \mu^2}, \quad (3)$$

such that the dependence on the frequency is intrinsically multi-valued in the complex plane. To treat this structure systematically, we introduce a uniformisation via the Zhukovsky transformation [24], which removes the square root and rewrites the problem in terms of a single complex parameter z . While the original formulation is quadratic in Ω , this mapping introduces both positive and negative powers of z , resulting in a seventh-degree polynomial eigenvalue problem, whose increased algebraic order reflects the nontrivial analytic structure of the massive dispersion relation. In this sense, the higher-degree formulation in the numerics is a direct consequence of uniformising the multi-valued analytic structure of the frequency domain.

From this analytical perspective, the wavenumber $k(\Omega)$ has branch points at $\Omega = \pm\mu$ in the complex Ω -plane, such that the relevant analytic structure is a two-sheeted Riemann surface. The Zhukovsky transformation resolves this structure by encoding the sheets in the reciprocal pair of z and $1/z$, while mapping the branch points to $z = \pm 1$. In this way, the analytic continuation required by the outgoing boundary condition at infinity is incorporated directly into the spectral formulation, thereby ensuring that the computed spectrum corresponds to the physically relevant analytic continuation.

Our approach is therefore novel in two respects: it extends the spectral method from a quadratic to a seventh-degree polynomial eigenvalue problem, and it provides a unified framework in which the complex-plane geometry,

Riemann-surface structure, and physical interpretation of the spectrum are treated consistently. To distinguish this approach from previous iterations of the spectral method techniques, we refer to this formulation as the “extended spectral method”.

The structure of this paper is as follows. We begin by outlining the eigenvalue problem of the massive scalar QNM propagating on a Schwarzschild background in Section II. In Section III, we present the explicit construction of the massive QNM eigenvalue problem, including a uniformisation of the multi-valued asymptotic wavenumber that recasts the spectral discretisation into a polynomial eigenvalue problem of higher degree, and describe how this formulation allows for a consistent extraction of physically relevant modes. In Section IV, we showcase our results, and discuss our findings in Section V. We summarise our conclusions and outlook in Section VI.

II. METRIC AND EQUATIONS OF MOTION

To introduce the QNM eigenvalue problem and clarify our conventions, we shall briefly review the massive scalar equation oscillating in the Schwarzschild background. In Planck units, where $c = G_N = \hbar = 1$, the Klein-Gordon equation for a massive scalar field in a curved geometry is expressed as

$$\frac{1}{\sqrt{-g}}\partial_\nu(g^{\mu\nu}\sqrt{-g}\partial_\mu\psi) - m^2\psi = 0. \quad (4)$$

In the Schwarzschild metric,

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2(d\vartheta^2 + \sin^2\vartheta d\varphi^2), \quad f(r) = 1 - \frac{2M}{r}, \quad (5)$$

such that the corresponding radial equation emerging from the separation of variables takes the form [8]

$$f(r)\frac{d}{dr}\left(f(r)\frac{d\psi}{dr}\right) + [\omega^2 - V(r)]\psi(r) = 0, \quad (6)$$

with the QNM effective potential,

$$V(r) = f(r)\left(\frac{L}{r^2} + \frac{2M}{r^3} + m^2\right), \quad L = \ell(\ell + 1). \quad (7)$$

In what follows, it is convenient to introduce the rescaling $x = r/(2M)$; we also make use of the dimensionless quantities $\Omega = M\omega$ and $\mu = Mm$. This allows us to rewrite the effective potential as

$$V(x) = \left(1 - \frac{1}{x}\right)\left(\frac{L}{4M^2x^2} + \frac{1}{4M^2x^3} + \left(\frac{\mu}{M}\right)^2\right). \quad (8)$$

We may then define $\tilde{V}(x) \equiv M^2V(x)$ to produce the potential in its fully dimensionless form,

$$\tilde{V}(x) = \left(1 - \frac{1}{x}\right)\left(\frac{L}{4x^2} + \frac{1}{4x^3} + \mu^2\right). \quad (9)$$

In Fig. 1, we plot this potential for $\ell = 2$ and increasing values of μ . Note that the $x = r/(2M)$ rescaling shifts the event horizon to $x = 1$. In the eikonal regime ($\ell \gg 1$), the effective potential is dominated by the angular momentum barrier L/x^2 , whose maximum coincides with the radius of the unstable circular null geodesic. Consequently, the potential is photon-sphere dominated in this regime, and the wave dynamics are governed by the properties of this orbit. In contrast, when $\mu \gg \ell$, the potential is dominated by the mass term μ^2 . We see explicitly in Fig. 1 that μ determines the asymptotic behaviour of the potential at infinity. As $L \sim \mu^2$, the narrow barrier is smoothed out and photon-sphere trapping fails. In that case, the resonances behave not as QNMs but rather as long-lived bound states.

We can determine explicitly for which parameter values the rescaled potential, Eq. (9), develops a barrier i.e. a local maximum followed by a local minimum for $x > 1$. To do so, we consider its derivative, where a short computation shows that

$$\tilde{V}'(x) = \frac{4\mu^2x^3 - 2Lx^2 + 3(L - 1)x + 4}{4x^5} = \frac{P(x)}{4x^5}. \quad (10)$$

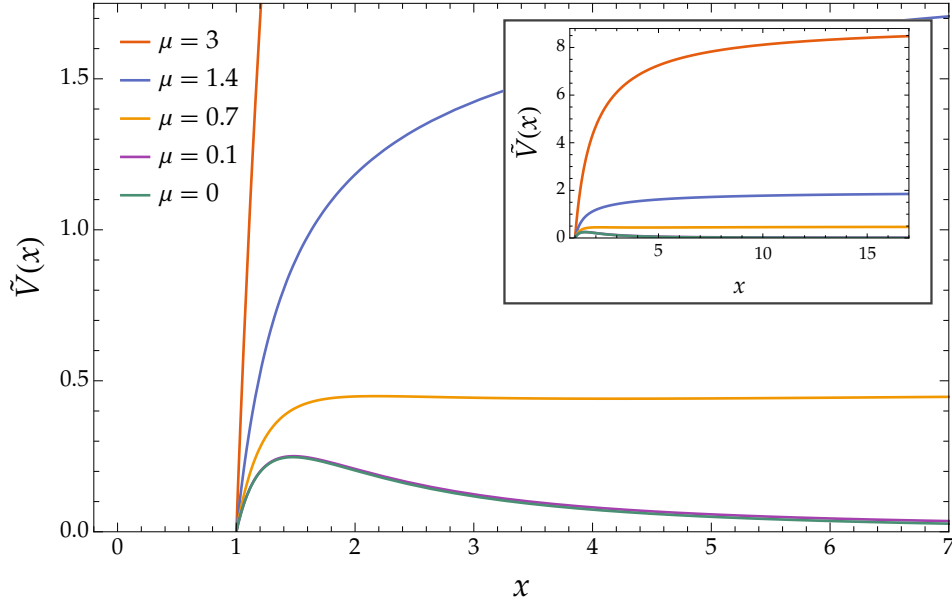


FIG. 1: Sketch of the effective QNM potential (Eq. (9)), where $\tilde{V}(x) \rightarrow \mu^2$ as $x \rightarrow \infty$. For the massless case, the locus of the peak in the barrier potential (the photon sphere) is at $x = 3/2$. As μ grows to 0.7 and beyond, however, $\tilde{V}(x)$ no longer manifests as a confining barrier potential.

Then the stationary points are given by the roots of the cubic,

$$P(x) = 4\mu^2 x^3 - 2Lx^2 + 3(L-1)x + 4 = 0. \quad (11)$$

For $\mu > 0$, the leading coefficient of $P(x)$ is positive. Moreover, $P(0) = 4 > 0$; for $0 < x \leq 1$, it can be easily checked that $P(x) \geq -2Lx^2 + 3(L-1)x + 4 > 0$, so that $P(x)$ has no roots in $(0, 1]$. Since $P(x) \rightarrow -\infty$ as $x \rightarrow -\infty$, and $P(x) \rightarrow +\infty$ as $x \rightarrow +\infty$, whenever $P(x)$ has three real roots, say x_0 , x_M , and x_m , these must be ordered as $x_0 < 0 < 1 < x_M < x_m$, where the two roots $x_M, x_m > 1$ correspond to a local maximum and a local minimum of the effective potential, respectively. Thus the potential exhibits the aforementioned maximum–minimum structure for $x > 1$ whenever Eq. (11) has three distinct real roots. This is the case when the discriminant is positive [25].

A straightforward calculation of the discriminant gives

$$\Delta(L, \mu) = 4 \left[-1728\mu^4 - 108(L-1)(L+1)^2\mu^2 + L^2(9L^2 + 14L + 9) \right]. \quad (12)$$

The condition $\Delta(L, \mu) > 0$ is equivalent to $1728\mu^4 + 108(L-1)(L+1)^2\mu^2 - L^2(9L^2 + 14L + 9) < 0$, which is a quadratic inequality in μ^2 with two real roots. The physically relevant $\mu > 0$ root is

$$\mu_c^2(L) = \frac{1}{288} \left[\sqrt{3}(3L^2 + 2L + 3)^{3/2} - 9(L-1)(L+1)^2 \right] > 0, \quad (13)$$

so that, for $L \geq 2$ corresponding to $\ell \geq 1$, the effective potential has a local maximum at $x_M > 1$ followed by a local minimum at $x_m > x_M$ when

$$0 < \mu^2 < \mu_c^2(L). \quad (14)$$

Let us consider an explicit example. For $\ell = 2$, we have seen in Fig. 1 that we lose a local maximum when $\mu > 0.7$. For $\ell = 2$, we have $L = 6$, and $\mu_c^2(L) \approx 0.5478$; $\mu^2 = (0.7)^2 = 0.49$ then satisfies Eq. (14) and the corresponding effective potential indeed develops a maximum and a subsequent minimum for $x > 1$. The corresponding allowed region in the (μ, L) -plane defined by Eq. (14) is displayed in Fig. 2, where the shaded area indicates the values of μ and L for which the potential develops a local maximum followed by a local minimum. There, a blue diamond corresponds to the aforementioned example, *viz.* $\mu = 0.5 < \mu_c(L = 6)$.

As a phenomenological aside, an interesting physical interpretation of the dimensionless mass parameter μ can be obtained by restoring SI units: we have

$$\mu = \frac{GM_{phys}}{c^2} \frac{m_{phys}c}{\hbar} = \frac{r_s}{2\lambda_C}, \quad (15)$$

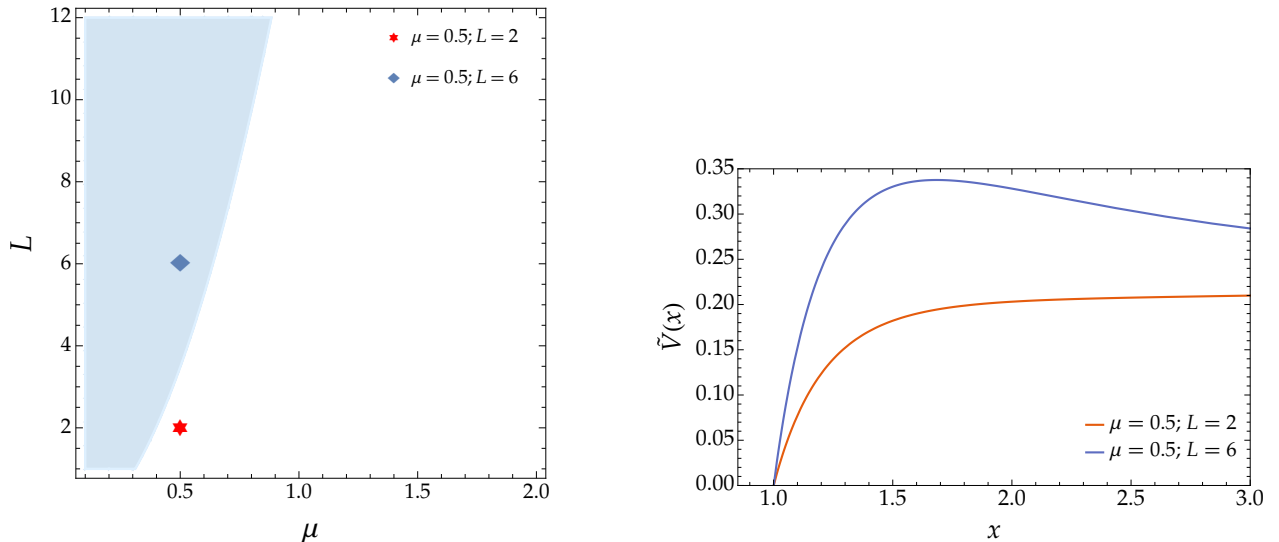


FIG. 2: *Left: Parameter region in the (μ, L) -plane where the effective potential Eq. (9) develops a local maximum followed by a local minimum for $x > 1$. The shaded area corresponds to the values of (μ, L) satisfying the constraint Eq. (14); specific points are marked for $\mu^2 > \mu_c^2(L=2)$ (red star) and $\mu^2 < \mu_c^2(L=6)$ (blue diamond), where $L = \ell(\ell+1)$. Right: Effective potential corresponding to the shaded (blue) and unshaded (red) regions, comparing the points corresponding to $\ell = 1$ and $\ell = 2$. Note that only the latter has a trapping potential.*

where $r_s = 2GM_{phys}/c^2$ is the Schwarzschild radius of the black hole and $\lambda_C = \hbar/(m_{phys}c)$ is the Compton wavelength of the scalar field oscillating in the vicinity of the black hole. Thus, μ quantifies the hierarchy between the length scale defined by the ADM mass of the black hole and the intrinsic quantum length scale of the field. That is, in the regime $\mu \ll 1$, the Compton wavelength is much larger than the horizon scale, whereas for $\mu \gg 1$, many Compton wavelengths fit inside the Schwarzschild radius and the field behaves effectively as a short-wavelength (geometric optics) probe. The intermediate regime, $\mu \sim \mathcal{O}(1)$, $r_s \sim \lambda_C$, is such that the Compton wavelength is comparable to the horizon scale. This is precisely the window in which the mass term induces a far-region potential well that supports quasibound states and very long-lived modes in Schwarzschild cases, and in Kerr superradiant growth (see Refs [26, 27] for further discussion on the ultralight-boson/superradiance motivation). To appreciate the mass scales of the field,

$$\mu \simeq 7.48 \times 10^9 \left(\frac{M_{BH}}{M_\odot} \right) \left(\frac{m}{eV} \right), \quad (16)$$

so $\mu \sim 1$ corresponds to $m \sim 1.3 \times 10^{-10} \text{ eV}$ for a one solar mass black hole and $m \sim 1.3 \times 10^{-16} \text{ eV}$ for $10^6 M_\odot$. By contrast, for Standard Model masses near astrophysical black holes, one typically has $r_s \gg \lambda_C$ and hence $\mu \gg 1$. The frequently used $0.5 \leq \mu \leq 1.5$ range [8] should therefore be read as a probe of ultralight fields whose Compton wavelengths are comparable to the black-hole size.

III. EXTENDED SPECTRAL METHOD

Before we consider the massive Schwarzschild problem, it is useful to recall the logic of the spectral construction used in the earlier formulation of the method [18–23]. After factoring out the known QNM asymptotics at the two boundaries and compactifying the radial domain to the finite interval $y \in [-1, 1]$, one is left with a regular unknown function, which is approximated by a truncated Chebyshev expansion,

$$\Phi_N(y) = \sum_{k=0}^N c_k T_k(y). \quad (17)$$

Collocating the differential equation at the Gauss-Lobatto points produces a finite-dimensional polynomial eigenvalue problem for the coefficient vector $\mathbf{c} = (c_0, \dots, c_N)^\top$. In the standard massless setting, the dependence on the frequency is a polynomial of degree two, so the discretised problem takes the form

$$P_2(\Omega)\mathbf{c} = (\Omega^2 A_2 + \Omega A_1 + A_0)\mathbf{c} = 0. \quad (18)$$

In this way, the previous version of the spectral method leads to a quadratic matrix pencil [28, 29]. Since the exact QNMs are not known *a priori*, the numerical uncertainty of a spectral computation is assessed *a posteriori* through stability under refinement. More precisely, if $\Omega_j^{(N)}$ denotes a mode computed with N basis functions, and $\Omega_j^{(N+\Delta N)}$ is the corresponding mode obtained after increasing the truncation order, we use the difference between these two values as an estimate of the error,

$$\varepsilon_j^{(N)} = \left| \Omega_j^{(N+\Delta N)} - \Omega_j^{(N)} \right|, \quad (19)$$

supplemented in practice by checks of stability under increased working precision. Roots that remain stable as N and the arithmetic precision are increased are interpreted as genuine modes, whereas strongly drifting roots are regarded as numerical artefacts. Note that we shall continue to rely on this criterion in Section IV, when comparing spectra computed with different truncation orders.

The massive case presented in this work preserves this general strategy, but alters the algebraic structure of the eigenvalue problem. As introduced in Section I, the asymptotic wavenumber is now defined as the multivalued quantity $k(\Omega) = \sqrt{\Omega^2 - \mu^2}$, so that the discretised operator is no longer a polynomial in Ω (due to the non-polynomial dependence introduced by the asymptotic wavenumber). The role of our “extended spectral method” is precisely to restore polynomiality by introducing a uniformising spectral parameter. The price one pays is that the quadratic pencil of the standard approach is replaced by a seventh-degree polynomial pencil. In this sense, the extended method is not a different numerical philosophy; instead, it represents a direct generalisation of the earlier spectral construction to a problem with non-trivial branch structure in the frequency plane.

With this overview in place, we now return to Eq. (6), which we can rewrite as

$$\frac{d^2\psi}{dx^2} + p(x)\frac{d\psi}{dx} + q(x)\psi(x) = 0 \quad (20)$$

with

$$p(x) = -\frac{1}{x} + \frac{1}{x-1}, \quad q(x) = \left(\frac{2x}{x-1} \right)^2 \left[\Omega^2 - \mu^2 + \frac{\mu^2}{x} - \frac{L}{4x^2} + \frac{L-1}{4x^3} + \frac{1}{4x^4} \right]. \quad (21)$$

Since p and q have poles of order one and two at $x = 1$, respectively, this point is a regular singular point of Eq. (20), according to Frobenius theory [30]. Hence, we can construct solutions of the form,

$$\psi(x) = (x-1)^\rho \sum_{\kappa=0}^{\infty} a_\kappa (x-1)^\kappa. \quad (22)$$

The leading behaviour at $x = 1$ is represented by the term $(x-1)^\rho$, where ρ is determined by the indicial equation,

$$\rho(\rho-1) + p_0\rho + q_0 = 0, \quad (23)$$

with

$$p_0 = \lim_{x \rightarrow 1} (x-1)p(x) = 1, \quad q_0 = \lim_{x \rightarrow 1} (x-1)^2 q(x) = 4\Omega^2. \quad (24)$$

The roots of Eq. (23) are $\rho_\pm = \pm 2i\Omega$, and the correct QNM boundary condition at the event horizon, $x = 1$, takes the form

$$\psi \xrightarrow{x \rightarrow 1^+} (x-1)^{-2i\Omega}. \quad (25)$$

The asymptotic behaviour of the solutions to Eq. (20) can be deduced using the method outlined in Ref. [31]. For this purpose, we start by observing that in the limit of $x \rightarrow +\infty$,

$$p(x) = \sum_{\kappa=0}^{\infty} \frac{f_\kappa}{x^\kappa} = \mathcal{O}\left(\frac{1}{x^2}\right), \quad q(x) = \sum_{\kappa=0}^{\infty} \frac{g_\kappa}{x^\kappa} = 4(\Omega^2 - \mu^2) + \frac{8\Omega^2 - 4\mu^2}{x} + \mathcal{O}\left(\frac{1}{x^2}\right). \quad (26)$$

Given that at least one of the coefficients f_0, g_0, g_1 is nonzero, a formal asymptotic solution to Eq. (20) is represented by [31]

$$\psi^{(\pm)}(x) = x^{\mu_\pm} e^{\lambda_\pm x} \sum_{\kappa=0}^{\infty} \frac{a_{\kappa,\pm}}{x^\kappa}, \quad (27)$$

where $\lambda_{\pm}$ and $\mu_{\pm}$ are the roots of the characteristic equations,

$$\lambda_{\pm}^2 + f_0 \lambda_{\pm} + g_0 = 0, \quad \mu_{\pm} = -\frac{f_1 \lambda_{\pm} + g_1}{f_0 + 2\lambda_{\pm}}. \quad (28)$$

Then a straightforward computation shows that

$$\lambda_{\pm} = \pm 2i\sqrt{\Omega^2 - \mu^2}, \quad \mu_{\pm} = \pm i\frac{2\Omega^2 - \mu^2}{\sqrt{\Omega^2 - \mu^2}}. \quad (29)$$

Hence, the QNM boundary condition at spatial infinity can be formulated as

$$\psi \xrightarrow{x \rightarrow +\infty} x^{i\mu_+} e^{i\lambda_+ x}. \quad (30)$$

In order to apply the spectral method, we first transform the radial function $\psi(x)$ into a new radial function $\Phi(x)$, such that the QNM boundary conditions, Eqs (25) and (30), are automatically implemented into the ansatz, and $\Phi_S(x)$ is regular at the event horizon and at spatial infinity. To this end, we consider the transformation

$$\psi(x) = x^{i\alpha}(x-1)^{-i\beta} e^{i\gamma(x-1)} \Phi(x), \quad (31)$$

$$\alpha = 2\Omega + \frac{2\Omega^2 - \mu^2}{\sqrt{\Omega^2 - \mu^2}}, \quad \beta = 2\Omega, \quad \gamma = 2\sqrt{\Omega^2 - \mu^2}, \quad \Omega \neq \mu. \quad (32)$$

If we substitute Eq. (31) into Eq. (20), we obtain the following ordinary differential equation (ODE) for the radial part, *viz.*

$$P_2(x)\Phi''(x) + P_1(x)\Phi'(x) + P_0(x)\Phi(x) = 0, \quad (33)$$

with

$$P_2(x) = 1, \quad (34)$$

$$P_1(x) = 2i\gamma + \frac{2i\alpha - 1}{x} + \frac{1 - 2i\beta}{x-1}, \quad (35)$$

$$P_0(x) = \frac{L+1-2\alpha(\beta+\gamma)-i(\alpha+\beta+\gamma)}{x} + \frac{(\beta+\gamma)^2 + 2\alpha\beta - L - 1 + i(\alpha+\beta+\gamma)}{x-1} + \frac{(1-i\alpha)^2}{x^2}. \quad (36)$$

Let us now introduce the transformation $x = 2/(1-y)$, which maps the points at infinity and the event horizon to $y = 1$ and $y = -1$, respectively. Then, Eq. (33) becomes

$$S_2(y)\ddot{\Phi}(y) + S_1(y)\dot{\Phi}(y) + S_0(y)\Phi(y) = 0, \quad (37)$$

where a dot denotes differentiation with respect to the new variable y , and

$$S_2(y) = \frac{(1-y)^4}{4}, \quad (38)$$

$$S_1(y) = -\frac{(1-y)^3}{2} + \frac{(1-y)^2}{2} \left[2i\gamma + (1-2i\beta)\frac{1-y}{1+y} + \frac{2i\alpha-1}{2}(1-y) \right], \quad (39)$$

$$S_0(y) = \frac{1}{2} [L+1-2\alpha(\beta+\gamma)-i(\alpha+\beta+\gamma)](1-y) + [(\beta+\gamma)^2 + 2\alpha\beta - L - 1 + i(\alpha+\beta+\gamma)] \frac{1-y}{1+y} + \frac{(1-i\alpha)^2}{4}(1-y)^2. \quad (40)$$

Note that we must also require that $\Phi(y)$ is regular at $y = \pm 1$.

Table I shows that the coefficients of the differential equation (37) share a common zero of order 2 at $y = 1$, while $y = -1$ is a pole of order 1 for the coefficients S_1 and S_0 . Therefore, to bring Eq. (37) into a form suitable for the spectral method, we first multiply it by $(1+y)/(1-y)^2$. Making use of the identities,

$$\beta^2 - \gamma^2 - 4\mu^2 = 0, \quad (\alpha - \beta)\gamma - \frac{\beta^2 + \gamma^2}{2} = 0, \quad (41)$$

TABLE I: Classification of the points $y = \pm 1$ for the relevant functions defined by Eqs (38), (39), and (40). Here, the abbreviations “z ord n ” and “p ord m ” stand for “zero of order n ” and “pole of order m ”, respectively.

y	$S_2(y)$	$S_1(y)$	$S_0(y)$
-1	+4	p ord 1	p ord 1
+1	z ord 4	z ord 2	z ord 2

Eq. (37) simplifies to the form

$$M_2(y)\ddot{\Phi}(y) + M_1(y)\dot{\Phi}(y) + M_0(y)\Phi(y) = 0, \quad (42)$$

where

$$M_2(y) = \frac{(1+y)(1-y)^2}{4}, \quad (43)$$

$$M_1(y) = i\gamma(1+y) + \frac{1-2i\beta}{2}(1-y) + \frac{1}{4} \left[i \frac{(\beta+\gamma)^2}{\gamma} - 3 \right] (1-y^2), \quad (44)$$

$$M_0(y) = -\frac{L+1}{2} + \frac{(\beta+\gamma)^3}{2\gamma} + i \frac{(\beta+\gamma)(\beta+3\gamma)}{4\gamma} + \frac{1}{4} \left[1 - \frac{(\beta+\gamma)^2}{2\gamma} \right]^2 (1+y). \quad (45)$$

We can then recover the correct asymptotic behaviour,

$$\lim_{y \rightarrow 1^-} M_2(y) = 0 = \lim_{y \rightarrow -1^+} M_2(y), \quad (46)$$

$$\lim_{y \rightarrow 1^-} M_1(y) = 2i\gamma, \quad \lim_{y \rightarrow -1^+} M_1(y) = 1 - 2i\beta, \quad (47)$$

$$\lim_{y \rightarrow 1^-} M_0(y) = -\frac{L}{2} - \frac{(\beta+\gamma)^3(\beta-3\gamma)}{8\gamma^2} + i \frac{\gamma^2 - \beta^2}{4\gamma}, \quad (48)$$

$$\lim_{y \rightarrow -1^+} M_0(y) = -\frac{L+1}{2} + \frac{(\beta+\gamma)^3}{2\gamma} + i \frac{(\beta+\gamma)(\beta+3\gamma)}{4\gamma}. \quad (49)$$

Because the coefficients γ and α depend explicitly on the non-polynomial quantity $\sqrt{\Omega^2 - \mu^2}$, Eq. (37) does not lead directly to a polynomial eigenvalue problem in Ω when discretised via the spectral method. To achieve the required polynomial dependence, we must introduce the transformation,

$$\Lambda = \Omega + \sqrt{\Omega^2 - \mu^2}. \quad (50)$$

This is a scaled version of the classical Joukowski (Zhukovsky) conformal mapping, $z \mapsto z + a^2/z$, originally introduced in the context of airfoil theory by Joukowski [24] and widely discussed in modern treatments of conformal mapping [32]. As discussed in Section I, this transformation provides the essential mechanism by which the multi-valued analytic structure of the dispersion relation is uniformised within the spectral formulation. By virtue of Eq. (50), we find that

$$\Omega = \frac{\Lambda^2 + \mu^2}{2\Lambda}, \quad \sqrt{\Omega^2 - \mu^2} = \frac{\Lambda^2 - \mu^2}{2\Lambda}. \quad (51)$$

A straightforward computation then demonstrates how this transforms the parameters of Eq. (31),

$$\alpha = \frac{2\Lambda^3}{\Lambda^2 - \mu^2}, \quad \beta = \frac{\Lambda^2 + \mu^2}{\Lambda}, \quad \gamma = \frac{\Lambda^2 - \mu^2}{\Lambda}. \quad (52)$$

As already mentioned, the extended spectral method incorporates the QNM boundary conditions via the QNM ansatz, Eq. (31), which correctly reproduces the required far-field behaviour. Indeed, we can verify that since

$$\alpha - \beta = \frac{2\Omega^2 - \mu^2}{\sqrt{\Omega^2 - \mu^2}}, \quad \gamma = 2\sqrt{\Omega^2 - \mu^2}, \quad (53)$$

TABLE II: Definitions of the coefficients L_{kj} appearing in Eq. (59), dependent on mass parameter μ and the angular momentum parametrisation $L = \ell(\ell + 1)$, as well as their corresponding behaviours at the endpoints of the interval $-1 \leq y \leq 1$.

(k, j)	$\lim_{y \rightarrow -1^+} L_{kj}$	L_{kj}	$\lim_{y \rightarrow 1^-} L_{kj}$
(0, 0)	0	0	0
(0, 1)	$-8\mu^6$	$-8\mu^6$	$-8\mu^6$
(0, 2)	0	0	0
(1, 0)	$2\mu^4(L + 1)$	$\mu^4(2L + 1 - y)$	$2\mu^4L$
(1, 1)	$-4\mu^4$	$-\mu^4(3y^2 - 2y - 1)$	0
(1, 2)	0	$-\mu^4(1 + y)(1 - y)^2$	0
(2, 0)	$4\mu^4$	$4\mu^4$	$4\mu^4$
(2, 1)	$8\mu^4$	$8\mu^4(y + 2)$	$24\mu^4$
(2, 2)	0	0	0
(3, 0)	$-4\mu^2(L + 1)$	$-2\mu^2(2L + 1 - y)$	$-4\mu^2L$
(3, 1)	$8\mu^2$	$2\mu^2(3y^2 - 2y - 1)$	0
(3, 2)	0	$2\mu^2(1 + y)(1 - y)^2$	0
(4, 0)	$-12\mu^2$	$4\mu^2(y - 2)$	$-4\mu^2$
(4, 1)	$8\mu^2$	$4\mu^2(y^2 - 4y - 3)$	$-24\mu^2$
(4, 2)	0	0	0
(5, 0)	$16\mu^2 + 2(L + 1)$	$16\mu^2 + 2L + 1 - y$	$8(2\mu^2 + L)$
(5, 1)	-4	$-(3y^2 - 2y - 1)$	0
(5, 2)	0	$-(1 + y)(1 - y)^2$	0
(6, 0)	8	$4(1 - y)$	0
(6, 1)	-8	$-4(y^2 - 2y - 1)$	8
(6, 2)	0	0	0
(7, 0)	-16	$4(y - 3)$	-8
(7, 1)	0	0	0
(7, 2)	0	0	0

we obtain, as $x \rightarrow +\infty$,

$$\psi(x) \sim x^{i(\alpha-\beta)} e^{i\gamma x},$$

up to an x -independent factor and inverse-power corrections. Thus, the asymptotic solution has the expected Coulomb-type power-law prefactor multiplied by $e^{i\gamma x}$. The sign of $\sqrt{\Omega^2 - \mu^2}$, equivalently of the asymptotic wavenumber,

$$k(\Omega) = \sqrt{\Omega^2 - \mu^2} = \frac{\gamma}{2}, \quad (54)$$

is therefore part of the boundary condition at spatial infinity. Note carefully that for propagative modes, the outgoing prescription is fixed by the branch satisfying $k(\Omega) \sim +\Omega$ as $|\Omega| \rightarrow \infty$. When the far-field behaviour becomes evanescent, the same branch is selected by analytic continuation from this outgoing sheet. The interpretation of the corresponding non-oscillatory modes is deferred to Section V. The Joukowski transformation of Eq. (50) uniformises the corresponding two-sheeted Riemann surface. Using Eq. (51), we may write

$$\Omega = \frac{\Lambda^2 + \mu^2}{2\Lambda}, \quad k(\Lambda) = \frac{\Lambda^2 - \mu^2}{2\Lambda}. \quad (55)$$

Hence, the two points on the Riemann surface lying over the same value of Ω are represented by the reciprocal pair,

$$\Lambda \longleftrightarrow \frac{\mu^2}{\Lambda}, \quad (56)$$

under which Ω is unchanged, whereas k and γ change sign. Equivalently, the branch points $\Omega = \pm\mu$ are mapped to $\Lambda = \pm\mu$, and the two sheets are separated in the Λ -plane by the circle $|\Lambda| = \mu$. The branch determined by $k(\Omega) \sim +\Omega$ corresponds to the exterior region $|\Lambda| > \mu$, while the reciprocal interior region $|\Lambda| < \mu$ represents the opposite sheet. Accordingly, the polynomial eigenvalue problem in Λ may return algebraic roots associated with either sheet.

In the numerical post-processing, we therefore retain the value of Λ itself and evaluate

$$k(\Lambda) = \frac{\Lambda^2 - \mu^2}{2\Lambda}, \quad \gamma(\Lambda) = \frac{\Lambda^2 - \mu^2}{\Lambda} = 2k(\Lambda), \quad (57)$$

which provides a direct diagnostic for assigning each computed root to the outgoing sheet or to its reciprocal counterpart. Recall that we retain Eq. (19) to estimate errors confirm numerical stability.

Finally, the Joukowski transformation, Eq. (50), allows us to obtain the purely polynomial spectral problem

$$L_0 + iL_1\Lambda + L_2\Lambda^2 + iL_3\Lambda^3 + L_4\Lambda^4 + iL_5\Lambda^5 + L_6\Lambda^6 + iL_7\Lambda^7 = 0 \quad (58)$$

with

$$L_k = L_{k0}(y)\Phi(y) + L_{k1}(y)\dot{\Phi}(y) + L_{k2}(y)\ddot{\Phi}(y) \quad (59)$$

for each $k \in \{0, 1, \dots, 7\}$. Moreover, in Table II, we have summarised the L_{kj} appearing in Eq. (59) and their limiting values at $y = \pm 1$. Once Λ is determined numerically, Ω can be recovered by means of the first formula appearing in Eq. (51). Since Ω is a rational function of Λ , no square root or branch choice is required when retrieving Ω from Λ .

The formulation above yields a well-defined polynomial eigenvalue problem of degree seven in the spectral parameter Λ , from which the physical frequencies Ω can be recovered unambiguously via Eq. (51). This reformulation makes explicit that the increased algebraic degree of the spectral problem is a direct consequence of uniformising the multi-valued dispersion relation, rather than an artefact of the discretisation. In the following section, we apply this extended spectral method to compute the QNMs of a massive scalar field in a Schwarzschild background, and analyse the resulting mode structure across both propagative and evanescent regimes.

IV. NUMERICAL RESULTS

We begin this section by validating the extended spectral method through a comparison with results from the literature computed using what we have referred to in this work as “traditional” techniques – specifically, the sixth-order WKB technique [33] and the CFM [4], where the latter is more reliable for the $\ell \sim n$ parameters explored in this comparison. To perform this validating step, we set the rescaled mass parameter μ in Eq. (58) to zero; in this limit, the master equation reduces to that of a massless scalar field propagating in the Schwarzschild background. The corresponding QNFs are displayed in Table III, which therefore serves as a benchmark test for the present numerical scheme. Note the evolution of the QNFs with the dimensionless parameters of ℓ and n : $\Re[\Omega]$ increases (decreases) with ℓ (n) while $\Im m[\Omega]$ decreases (increases) with ℓ (n). We find that at order $\mathcal{O}(10^{-4})$, the spectral methods remain in agreement for $N = 40$ and $N = 200$, demonstrating that at this order, greater precision cannot be achieved by increasing the iterative depth. The excellent agreement with established results in the literature, particularly with the CFM, validates both the implementation and the accuracy of our extended spectral method.

We proceed with our exploration of the parameter space in Tables IV, V, VI, and VII, where we list our QNF results for the massive scalar field in the Schwarzschild background for intermediate values of the rescaled mass parameter, i.e. $0 < \mu < 1$. Specifically, we consider $\mu = 0.1, 0.2$ in Tables IV and V, and $\mu = 0.7, 0.8$ in Tables VI and VII; for these choices of μ , we compute the QNFs for the first few multipoles and overtones. In Tables IV and VI, we observe that $\Re[\Omega]$ ($\Im m[\Omega]$) increases (decreases) with μ , in accordance with the expected massive QNM behaviour. We introduce the $\chi^2 = \Re[\Omega]^2 - \mu^2$ parameter that distinguishes between evanescent and propagative modes: as described in the introduction, when $\Re[\Omega]^2 < \mu^2$, the QNMs behave as evanescent modes. Upon comparing Tables IV and VI, we see that evanescent modes dominate the latter; in the former, only for the least energetic ($\ell = 0$) and damped ($n \neq 0$) QNFs do we find that $\chi^2 < 0$, whereas in the latter, only evanescent modes appear. This behaviour follows from the structure of the QNM effective potential, whose local maximum decreases relative to the asymptotic value μ^2 as the mass parameter increases, and disappears once μ^2 exceeds the barrier peak.

In addition to the standard damped propagative and evanescent modes, we identify families of purely imaginary modes, with approximately evenly spaced imaginary parts. For $\mu = 0.1, 0.2$ and $\mu = 0.7, 0.8$, these are displayed in

TABLE III: QNFs of massless scalar perturbations ($\mu = 0$) of the Schwarzschild black hole. We compare our extended spectral method results with $N = 200$ Chebyshev polynomials (final column) against the CFM of Ref. [4] (third column), the sixth-order WKB computations of Ref. [33] (fourth column), and Ref. [34]’s spectral method results computed using $N = 40$ Chebyshev polynomials. Entries labelled “N/A” indicate unavailable data.

ℓ	n	Ω_{CFM} [4]	Ω_{WKB} [33]	$\Omega_{N=40}$ [34]	$\Omega_{N=200}$
0	0	$0.1105 - 0.1049i$	$0.1105 - 0.1008i$	$0.1105 - 0.1049i$	$0.1105 - 0.1049i$
	1	$0.0861 - 0.3481i$	$0.0890 - 0.3446i$	N/A	$0.0861 - 0.3481i$
	2	N/A	$0.1918 - 0.4765i$	N/A	$0.0757 - 0.6011i$
1	0	$0.2929 - 0.0977i$	$0.2929 - 0.0978i$	$0.2929 - 0.0977i$	$0.2929 - 0.0977i$
	1	$0.2645 - 0.3063i$	$0.2645 - 0.3065i$	$0.2645 - 0.3063i$	$0.2644 - 0.3063i$
	2	$0.2295 - 0.5401i$	$0.2310 - 0.5422i$	N/A	$0.2295 - 0.5401i$
	3	$0.2033 - 0.7883i$	$0.2221 - 0.7952i$	N/A	$0.2033 - 0.7883i$
2	0	$0.4836 - 0.0968i$	$0.4836 - 0.0968i$	$0.4836 - 0.0968i$	$0.4836 - 0.0968i$
	1	$0.4639 - 0.2956i$	$0.4638 - 0.2956i$	$0.4639 - 0.2956i$	$0.4639 - 0.2956i$
	2	$0.4305 - 0.5086i$	$0.4304 - 0.5087i$	$0.4305 - 0.5086i$	$0.4305 - 0.5086i$
	3	$0.3939 - 0.7381i$	$0.3932 - 0.7399i$	N/A	$0.3939 - 0.7381i$

Tables V and VII, respectively. The magnitude of the imaginary component increases monotonically with overtone number, while the spacing between successive modes remains approximately constant. This feature has not been reported in the literature for massive scalar fields propagating on a Schwarzschild background, beyond the claims of spurious modes in, for example, Ref. [15]. The numerical stability demonstrated in Fig. 3, however, suggests that these are mathematically valid. Furthermore, the robustness of these modes is assessed using the spectral convergence criterion of Eq. (19), together with stability checks under increasing truncation order and numerical precision, which allows us to distinguish them as genuine modes rather than numerical artefacts.

Recall also that purely imaginary spectra have been reported in the literature. For example, within Schwarzschild-de Sitter spacetimes, the QNF spectrum splits into two distinct branches, one of which presents as purely imaginary in the $\mu M \ll 1$ regime [35]. On the other hand, for massive scalar fields on a Schwarzschild background, such structures have not been systematically resolved. Existing studies (e.g. Ref. [8]) present only a limited subset of overtones and do not provide sufficient numerical detail for quantitative comparison. We return to the interpretation of these purely imaginary modes in Section V.

Finally, in Tables VIII, IX, X, XI, and XII, we investigate the critical regime of the mass parameter, $1.3 \leq \mu \leq 2.2$. Specifically, we consider $\mu \in \{1.30, 1.40, 1.42\}$ in Tables VIII and IX, and $\mu \in \{1.45, 1.50, 2.20\}$ in Tables X and XI; for these choices of μ , we compute the QNFs for the first few multipoles and overtones. We observe that Tables VIII and X are dominated by evanescent modes: for all but the $n = \ell = 0$ set of QNFs, $\Re[\Omega]^2 < \mu^2$, propagative QNMs do not arise. For this reason, we list the QNMs that can be captured by our extended spectral method for $\mu \in \{1.30, 1.40, 1.42, 1.45, 1.50, 2.20\}$ in Table XII.

In Tables VIII and X, we observe non-monotonic behaviour for increasing n in the evanescent modes for $\ell = 0$ and $\mu = 1.30, 1.40, 1.45, 1.50, 2.20$; for $\ell = 1$ and $\mu = 1.42$; for $\ell = 2$ and $\mu = 1.50, 2.20$. For the heavy mass parameter examined in Table XII, we observe that $\Re[\Omega]$ increases with μ for $\ell = n = 0$ – unlike the evanescent modes of Tables VIII and X, where $\Re[\Omega]$ decreases with μ for fixed ℓ and n .

As before, we identify families of purely imaginary modes, exhibiting monotonic behaviour with approximately equal spacing between subsequent overtones in Tables IX and XI, in addition to the evanescent modes studied in Tables VIII and X, respectively. Unlike previous cases, however, within Tables IX and XI, we find that the extended spectral method does not produce overdamped modes for $\ell = 0$ when $\mu = 1.40, 1.42$ and for $\ell = 2$ when $\mu = 2.20$.

TABLE IV: QNFs for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\chi^2 = \Re[\Omega]^2 - \mu^2$ ($M = 1$). “N/A” indicates data not available.

μ	ℓ	n	Ω	χ^2	μ	ℓ	n	Ω	χ^2
0.1	0	0	$0.1124 - 0.0968i$	0.0026	0.2	0	0	$0.1162 - 0.0754i$	-0.0265
		1	$0.0849 - 0.3467i$	-0.0028			1	$0.0819 - 0.3427i$	-0.0333
		2	$0.0752 - 0.6006i$	-0.0043			2	$0.0736 - 0.5992i$	-0.0346
	1	0	$0.2974 - 0.0950i$	0.0784		1	0	$0.3110 - 0.0866i$	0.0567
		1	$0.2647 - 0.3029i$	0.0601			1	$0.2651 - 0.2928i$	0.0303
		2	$0.2287 - 0.5386i$	0.0423			2	$0.2261 - 0.5341i$	0.0111
		3	$0.2026 - 0.7876i$	0.0310			3	$0.2005 - 0.7857i$	0.0002
		4	$0.1846 - 1.0404i$	0.0241			4	$0.1832 - 1.0394i$	-0.0064
		5	$0.1717 - 1.2939i$	0.0195			5	$0.1707 - 1.2933i$	-0.0109
	2	0	$0.4868 - 0.0957i$	0.2270		2	0	$0.4963 - 0.0924i$	0.2063
		1	$0.4655 - 0.2933i$	0.2067			1	$0.4703 - 0.2862i$	0.1812
		2	$0.4307 - 0.5065i$	0.1755			2	$0.4310 - 0.5002i$	0.1458
		3	$0.3934 - 0.7368i$	0.1448			3	$0.3919 - 0.7328i$	0.1136
		4	$0.3607 - 0.9792i$	0.1201			4	$0.3591 - 0.9769i$	0.0890
		5	$0.3344 - 1.2279i$	0.1018			5	$0.3331 - 1.2266i$	0.0710

V. DISCUSSION

Our results demonstrate several non-trivial features of the massive scalar QNM spectrum in a Schwarzschild background. Most notably, we identify the presence of purely imaginary modes in specific regions of the parameter space, which remain sensitive to the field mass μ even at moderately large overtones, $n \sim \mathcal{O}(10)$. These modes are captured robustly by the extended spectral method, which directly solves the global boundary value problem and remains reliable in regimes where asymptotic or WKB-type approximations are known to break down. The boundary conditions at the event horizon and at spatial infinity are implemented through the ansatz in Eq. (31), which incorporates the massive dispersion relation via the asymptotic wavenumber $k(\Omega) = \sqrt{\Omega^2 - \mu^2}$, as described in Section III. In particular, the outgoing branch is selected by $k(\Omega) \sim +\Omega$ as $|\Omega| \rightarrow \infty$ for propagating modes, while evanescent modes are obtained by analytic continuation from this outgoing sheet. The Joukowski transformation of Eq. (50) uniformises the associated two-sheeted Riemann surface, ensuring that all modes – including non-oscillatory, purely imaginary solutions – are treated within a single, consistent framework.

As illustrated in Section II, introducing a mass term into the QNM eigenvalue problem modifies the far-region structure of the effective potential, producing a nonzero asymptote and, in some cases, a shallow well. As a consequence, the QNM spectrum depends on the global potential and can carry a “hair” associated with the field mass. The influence of scalar field mass, as well as angular momentum and overtone numbers, on the QNF spectrum is recorded in Section IV. There, we observe the evolution of the massive QNFs with the dimensionless parameters of ℓ and n : when $\mu = 0$, $\Re[\Omega]$ increases (decreases) with ℓ (n) while $\Im m[\Omega]$ decreases (increases) with ℓ (n). For small and intermediate masses $\mu < 1$, $\Re[\Omega]$ ($\Im m[\Omega]$) increases (decreases) with μ , in accordance with the expected massive QNM behaviour. Evanescent modes dominate for heavier mass values: for the remaining QNMs examined in Table XII, we observe that $\Re[\Omega]$ increases with μ for $\ell = n = 0$ – unlike the evanescent modes of Tables VIII and X, where $\Re[\Omega]$ decreases with μ for fixed ℓ and n .

In this context, it is instructive to revisit earlier claims in the literature regarding the behaviour of massive scalar modes in asymptotic limits. Within the QNM literature, the large- n regime has attracted particular attention because of its purported connection to black hole microphysics. Specifically, Hod noted that $\Re[\omega] \sim \ln[3]/(8\pi M)$ as $n \rightarrow \infty$ in a Schwarzschild background [36], which he interpreted as a potential universal signature linked to black-hole entropy and the spacing of area eigenvalues [37]. This observation motivated a series of analytic derivations, most notably by Motl [38], who obtained the $\ln[3]$ result for massless scalar and gravitational perturbations via monodromy methods applied to the radial equation in the complex plane. This result was later used in Ref. [8] to argue that the same

TABLE V: The purely imaginary modes for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The corresponding results are obtained through our extended spectral method, utilising 200 polynomials with a precision of 200 digits. Here, $\Delta\Omega = \Omega_n - \Omega_{n+1}$ and $\chi^2 = -\mu^2$.

μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$
0.1	0	0	0.0000 – 19.36677i	N/A	0.2	0	0	0.0000 – 19.36676i	N/A
		1	0.0000 – 19.61682i	0.25006i			1	0.0000 – 19.61681i	0.25005i
		2	0.0000 – 19.86681i	0.24999i			2	0.0000 – 19.86680i	0.24999i
		3	0.0000 – 20.11691i	0.25010i			3	0.0000 – 20.11691i	0.25010i
		4	0.0000 – 20.36694i	0.25003i			4	0.0000 – 20.36693i	0.25002i
		5	0.0000 – 20.61699i	0.25005i			5	0.0000 – 20.61698i	0.25005i
		6	0.0000 – 20.86704i	0.25005i			6	0.0000 – 20.86704i	0.25005i
		7	0.0000 – 21.11709i	0.25004i			7	0.0000 – 21.11708i	0.25004i
	1	0	0.0000 – 20.32006i	0.25028i		1	0	0.0000 – 20.32005i	0.25028i
		1	0.0000 – 20.57037i	0.25031i			1	0.0000 – 20.57036i	0.25031i
		2	0.0000 – 20.82070i	0.25033i			2	0.0000 – 20.82070i	0.25033i
		3	0.0000 – 21.07099i	0.25029i			3	0.0000 – 21.07098i	0.25029i
		4	0.0000 – 21.32130i	0.25031i			4	0.0000 – 21.32129i	0.25031i
		5	0.0000 – 21.57160i	0.25030i			5	0.0000 – 21.57159i	0.25030i
		6	0.0000 – 21.82189i	0.25029i			6	0.0000 – 21.82188i	0.25029i
		7	0.0000 – 22.07218i	0.25029i			7	0.0000 – 22.07217i	0.25029i
	2	0	0.0000 – 20.24222i	0.25071i		2	0	0.0000 – 20.24221i	0.25071i
		1	0.0000 – 20.49275i	0.25053i			1	0.0000 – 20.49274i	0.25053i
		2	0.0000 – 20.74348i	0.25073i			2	0.0000 – 20.74347i	0.25073i
		3	0.0000 – 20.99408i	0.25060i			3	0.0000 – 20.99407i	0.25060i
		4	0.0000 – 21.24470i	0.25062i			4	0.0000 – 21.24469i	0.25062i
		5	0.0000 – 21.49533i	0.25062i			5	0.0000 – 21.49531i	0.25063i
		6	0.0000 – 21.74592i	0.25059i			6	0.0000 – 21.74591i	0.25059i
		7	0.0000 – 21.99652i	0.25060i			7	0.0000 – 21.99651i	0.25060i

asymptotic behaviour persists for massive scalar fields. The reasoning there proceeds by considering the Leaver-type three-term recurrence relation for the massive scalar field and expanding the coefficients in powers of $\sqrt{\omega^2 - \mu^2}$ in the limit of large $|\omega|$. At leading order, the recurrence relation reduces to that of the massless problem, up to a shift in the effective multipole number. Since the high-overtone asymptotics in the massless case are known to be independent of ℓ ; this reduction is then taken to imply that the $\ln[3]/(8\pi M)$ behaviour continues to hold for any finite mass $\mu > 0$. In effect, this argument relies on importing the analytic structure of the massless problem into the massive case via an expansion that is formally valid only in the large-frequency regime.

The results presented here indicate that this extrapolation must be treated and interpreted with considerable caution. In our computations, we do not observe that the real part of the frequency approaches $\ln[3]/(8\pi M)$ as n increases for $\mu \neq 0$. The absence of this behaviour highlights a limitation of the analytic continuation procedure itself. The massive problem is not continuously connected to the massless one at the level of global boundary conditions: the non-vanishing asymptotic value of the effective potential alters the analytic structure of the radial equation and, in particular, the topology of the associated Stokes lines. As a result, arguments that rely on the vanishing of the potential at infinity (which is central to the monodromy technique of Ref. [38]) do not straightforwardly generalise to the massive case.

This observation has several important implications. Firstly, it supports the interpretation that highly-damped QNMs remain sensitive to the full structure of the effective potential, including its far-region behaviour, and are not determined solely by near-horizon physics. Secondly, it emphasises the limited domain of applicability of ana-

TABLE VI: QNMs (Ω) and evanescent modes (Ω_e) for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\Omega = M\omega$ denotes the dimensionless QNF, n the overtone number, and $\chi^2 = \Re[\Omega]^2 - \mu^2$ ($M = 1$). “N/A” indicates data not available.

μ	ℓ	n	Ω	Ω_e	χ^2	μ	ℓ	n	Ω	Ω_e	χ^2
0.7	0	0	N/A	$0.6954 - 0.0019i$	-0.0064	0.8	0	0	N/A	$0.7935 - 0.0032i$	-0.1804
		1	N/A	$0.6937 - 0.0032i$	-0.0088			1	N/A	$0.7913 - 0.0053i$	-0.1838
		2	N/A	$0.6910 - 0.0588i$	-0.0125			2	N/A	$0.7879 - 0.0093i$	-0.1892
		3	N/A	$0.6864 - 0.0120i$	-0.0189			3	N/A	$0.7824 - 0.0184i$	-0.1979
		4	N/A	$0.6787 - 0.0295i$	-0.0294			4	N/A	$0.7745 - 0.0424i$	-0.2101
		5	N/A	$0.6673 - 0.1046i$	-0.0447			5	N/A	$0.7682 - 0.1392i$	-0.2199
	1	0	N/A	$0.6775 - 0.1286i$	-0.0310		1	0	N/A	$0.7824 - 0.0102i$	-0.1979
		1	N/A	$0.6508 - 0.0361i$	-0.0665			1	N/A	$0.7704 - 0.0222i$	-0.2165
		2	N/A	$0.2011 - 0.4800i$	-0.4496			2	N/A	$0.7393 - 0.0595i$	-0.2634
		3	N/A	$0.1792 - 0.7571i$	-0.4579			3	N/A	$0.1970 - 0.4634i$	-0.7712
		4	N/A	$0.1669 - 1.0225i$	-0.4621			4	N/A	$0.1747 - 0.7475i$	-0.7795
		5	N/A	$0.1582 - 1.2823i$	-0.4650			5	N/A	$0.1629 - 1.0163i$	-0.7835
	2	0	N/A	$0.2978 - 1.4655i$	-0.4013		2	0	N/A	$0.2937 - 1.4610i$	-0.7237
		1	N/A	$0.2839 - 1.7225i$	-0.4094			1	N/A	$0.2804 - 1.7192i$	-0.7314
		2	N/A	$0.2725 - 1.9784i$	-0.4157			2	N/A	$0.2694 - 1.9759i$	-0.7374
		3	N/A	$0.2625 - 2.2334i$	-0.4211			3	N/A	$0.2600 - 2.2315i$	-0.7424
		4	N/A	$0.2540 - 2.4879i$	-0.4255			4	N/A	$0.2519 - 2.4863i$	-0.7465
		5	N/A	$0.2466 - 2.7418i$	-0.4292			5	N/A	$0.2447 - 2.7405i$	-0.7501

lytic continuations that connect the massless and massive problems. While such continuations can reproduce certain leading-order features, we reiterate that they may fail to capture changes in the spectrum that arise from modifications of the boundary conditions at infinity. The example discussed here illustrates how an apparently controlled approximation – namely, the reduction of the recurrence relations to their massless form – can obscure essential aspects of the problem. In this sense, the absence of the $\ln[3]/(8\pi M)$ behaviour in our results highlights a deeper structural distinction between the two cases rather than a simple quantitative discrepancy with earlier work, and provides a concrete illustration of how seemingly controlled approximations can lead to misleading conclusions if they are assumed to apply outside of their domain of validity.

The discrepancies observed in the quasis resonant regime further reinforce this point. While purely real modes (quasis resonances) were first predicted in Ref. [9] and later studied in Ref. [8], our results demonstrate significant quantitative differences in their frequencies. For example, for $\ell = 0$ and $\mu = 1.42$ (Table VIII), we obtain $\Re[\Omega] = 1.4424$ (with $\Re[\Omega_e] = 2.1531$), in contrast to the much smaller value inferred from the figures of Ref. [8]. For $\ell = 1$ and $\mu = 2.20$ (Table X), we find $\Re[\Omega] = 2.3906$ (with $\Re[\Omega_e] = 1.3953$), which is in closer qualitative agreement with previous results.

We reiterate that these discrepancies can be traced back to the same underlying cause: the increasing importance of the far-region structure of the effective potential in the quasis resonant regime. As the potential develops a shallow well and a nonzero asymptote, the spectrum becomes highly sensitive to subleading contributions that are not captured by large-frequency expansions or truncated recurrence relations. As emphasised in Ref. [13], the applicability of a given computational method must therefore be assessed carefully in this regime.

In contrast, the extended spectral method employed here does not rely on asymptotic matching or perturbative expansions. By solving the full boundary value problem with the correct analytic structure built in from the outset, it is able to resolve the spectrum across different regimes with controlled numerical precision. This includes not only the quasis resonant modes but also the purely imaginary branches identified in this work, which to our knowledge have not been reported previously. Taken together, these results demonstrate that global methods of this type are essential for capturing the full complexity of QNM spectra in the presence of massive fields, and for avoiding artefacts associated

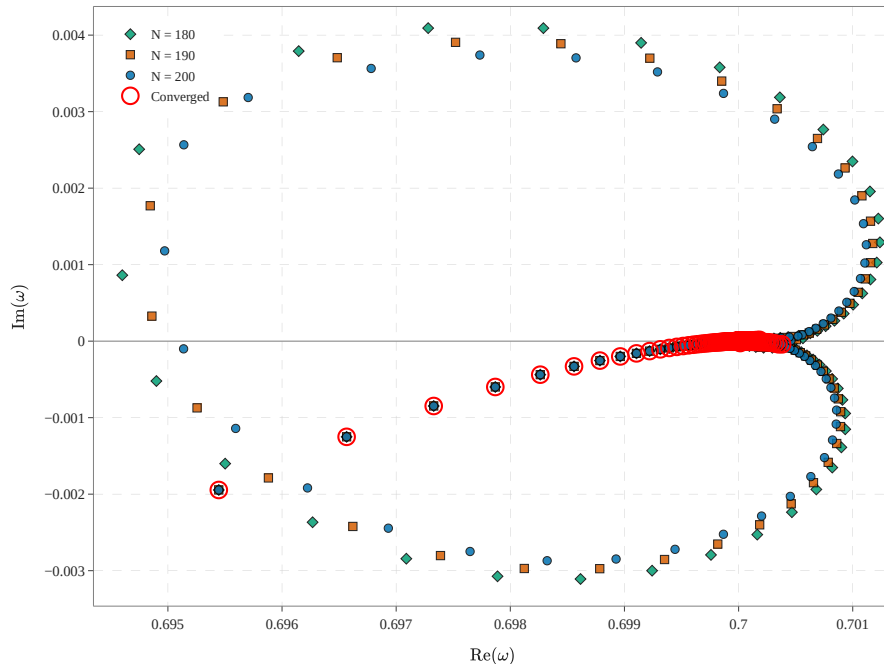


FIG. 3: Discrete evanescent modes for $\mu = 0.7$ and $\ell = 0$, computed by Chebyshev collocation with $N = 180, 190, 200$. The red circles indicate the roots that remain stable as N varies. Their pronounced clustering near $\omega = \mu + 0i$ is consistent with the collocation signature of an isolated defective, or nearly defective, QNM [28, 29]: at finite truncation order, one does not expect exact defectivity, but rather a cluster of nearby discrete eigenvalues whose total algebraic multiplicity approximates that of the underlying continuous spectral point. Here N denotes the number of Chebyshev polynomials.

with incomplete analytic continuations.

VI. CONCLUSIONS AND OUTLOOK

In this work, we revisited the case of massive scalar QNMs propagating on a Schwarzschild background. Traditional computational techniques applied to the corresponding QNF spectra are typically restricted to small field masses and/or regimes in which $\ell \gg n$. To overcome these limitations, (pseudo)spectral methods have emerged as powerful alternatives. Here, we demonstrated the flexibility of such approaches by extending the spectral method introduced in Refs [18–23]. In particular, we showed explicitly how the physically correct boundary conditions for the massive QNM problem must be incorporated, thereby reformulating the method into what we refer to as the “extended spectral method”.

We validated this framework by comparing it with existing results in Table III, finding good agreement in the regimes where other methods are expected to be reliable. Beyond this, the method proves to be more robust, as it allows us to probe regions of parameter space that are typically inaccessible, including $\ell < n$ and $\Re[\omega^2] < \mu^2$. In practice, this enabled us to compute several overtones and harmonics across an extended range of field masses, including regimes in which the effective potential develops features such as far-region wells and non-vanishing asymptotic values.

A particularly interesting outcome of this work is the emergence of a branch of purely imaginary modes as the number of Chebyshev polynomials N is increased (see Fig. 3). We observed that this imaginary branch exhibited a high degree of structure, with an approximately constant spacing in $\Im m[\Omega]$, and remained stable under variations of the physical parameters. Such behaviour is inconsistent with expectations from spectral pollution, which is typically irregular and unstable, and therefore suggests that these modes represent a genuine feature of the spectrum. In this sense, the extended spectral method appears to capture aspects of the QNM structure that are not accessible to more traditional approaches.

Furthermore, Fig. 3 illustrates a phenomenon that fits naturally into the “defective cluster theory” [28, 29]. As the truncation order N increases, a group of numerically stable roots accumulates near $\omega = \mu + 0i$. From a single

TABLE VII: *Purely imaginary QNMs for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The corresponding results are obtained through our extended spectral method, utilising 200 polynomials with a precision of 200 digits. In this context, $\Omega = M\omega$ and n represent the dimensionless frequency and the corresponding overtone, respectively, while $\Delta\Omega = \Omega_n - \Omega_{n+1}$. Here, we find that $\chi^2 = -\mu^2$.*

μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$
0.7	0	0	0.0000 – 19.36665i	N/A	0.8	0	0	0.0000 – 19.36661i	N/A
		1	0.0000 – 19.61669i	0.25005i			1	0.0000 – 19.61665i	0.25004i
		2	0.0000 – 19.86669i	0.24999i			2	0.0000 – 19.86665i	0.25000i
		3	0.0000 – 20.11679i	0.25011i			3	0.0000 – 20.11675i	0.25011i
		4	0.0000 – 20.36682i	0.25002i			4	0.0000 – 20.36678i	0.25002i
		5	0.0000 – 20.61687i	0.25006i			5	0.0000 – 20.61684i	0.25006i
		6	0.0000 – 20.86693i	0.25005i			6	0.0000 – 20.86689i	0.25005i
		7	0.0000 – 21.11697i	0.25004i			7	0.0000 – 21.11694i	0.25004i
	1	0	0.0000 – 20.31992i	0.25027i		1	0	0.0000 – 20.31988i	0.25027i
		1	0.0000 – 20.57023i	0.25032i			1	0.0000 – 20.57019i	0.25032i
		2	0.0000 – 20.82057i	0.25034i			2	0.0000 – 20.82053i	0.25034i
		3	0.0000 – 21.07086i	0.25029i			3	0.0000 – 21.07082i	0.25030i
		4	0.0000 – 21.32118i	0.25031i			4	0.0000 – 21.32114i	0.25031i
		5	0.0000 – 21.57147i	0.25030i			5	0.0000 – 21.57144i	0.25030i
		6	0.0000 – 21.82177i	0.25029i			6	0.0000 – 21.82173i	0.25029i
		7	0.0000 – 22.07206i	0.25029i			7	0.0000 – 22.07202i	0.25029i
	2	0	0.0000 – 20.24202i	0.25069i		2	0	0.0000 – 20.24196i	0.25069i
		1	0.0000 – 20.49256i	0.25053i			1	0.0000 – 20.49249i	0.25053i
		2	0.0000 – 20.74330i	0.25074i			2	0.0000 – 20.74324i	0.25074i
		3	0.0000 – 20.99389i	0.25060i			3	0.0000 – 20.99383i	0.25060i
		4	0.0000 – 21.24452i	0.25063i			4	0.0000 – 21.24446i	0.25063i
		5	0.0000 – 21.49515i	0.25063i			5	0.0000 – 21.49509i	0.25063i
		6	0.0000 – 21.74574i	0.25060i			6	0.0000 – 21.74569i	0.25060i
		7	0.0000 – 21.99635i	0.25060i			7	0.0000 – 21.99629i	0.25060i

discrete spectrum, one cannot rigorously distinguish between a genuinely “defective” continuous QNM and several tightly packed semi-simple modes. Nevertheless, the observed behaviour is exactly what one expects if the underlying continuous problem contains an isolated defective, or nearly defective, spectral point: finite-dimensional collocation does not preserve exact defectivity, but typically resolves it into a nearby cluster of simple roots, whose total algebraic multiplicity is the relevant invariant and whose diameter is controlled by the defective splitting theory. In this sense, Fig. 3 can be interpreted as numerical evidence for a defective local spectral component rather than for a single exceptionally ill-conditioned root.

Our results also reinforce the notion that the asymptotic QNM spectrum is sensitive to the global structure of the effective potential. However, the absence of the $\ln[3]/(8\pi M)$ term in the highly-damped regime for massive fields provides further evidence against the universality of this term, and instead points toward a more nuanced picture in which the asymptotic behaviour depends on whether the potential decays to zero or approaches a constant at spatial infinity. This highlights the importance of far-region physics in determining the spectrum of highly-damped QNMs, even for modes that are often heuristically associated with near-horizon dynamics.

This extended method would have ramifications for other scenarios, such as for massive scalar fields (possibly with nonzero scalar field charge) in charged and/or rotating black-hole backgrounds, excluding or including a cosmological constant. In these cases, the additional structure in the effective potential would be expected to enrich further the QNM spectrum. Similarly, scenarios where there is an interplay between superradiance, mass terms, and asymptotic

TABLE VIII: *Evanescent modes (Ω_e) for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\Omega = M\omega$ denotes the dimensionless QNF, n the overtone number, and $\chi^2 = \Re[\omega]^2 - \mu^2$ (with $M = 1$). “N/A” indicates data not available.*

μ	ℓ	n	Ω_e	χ^2	μ	ℓ	n	Ω_e	χ^2	μ	ℓ	n	Ω_e	χ^2
1.30	0	0	1.4289-0.0174i	-0.0607	1.40	0	0	1.4774-0.0194i	-0.0673	1.42	0	0	2.1531-0.0620i	-0.2042
		1	1.4246-0.0247i	-0.0730			1	1.4729-0.0274i	-0.0806			1	2.1481-0.0826i	-0.2257
		2	1.4195-0.0365i	-0.0875			2	1.4676-0.0403i	-0.0962			2	2.1441-0.1131i	-0.2428
		3	1.4138-0.0564i	-0.1037			3	1.4620-0.0618i	-0.1126			3	2.1439-0.1603i	-0.2437
		4	1.4095-0.0931i	-0.1158			4	1.4585-0.1012i	-0.1228			4	2.1542-0.2393i	-0.1994
		5	1.4151-0.1716i	-0.1000			5	1.4659-0.1843i	-0.1011			5	2.1984-0.3935i	-0.0070
	1	0	1.4242-0.0203i	-0.0742		1	0	1.4723-0.0228i	-0.0823		1	0	2.1446-0.0746i	-0.2407
		1	1.4180-0.0298i	-0.0918			1	1.4659-0.0333i	-0.1011			1	2.1378-0.1020i	-0.2698
		2	1.4099-0.0459i	-0.1147			2	1.4577-0.0508i	-0.1251			2	2.1319-0.1442i	-0.2950
		3	1.3995-0.0750i	-0.1439			3	1.4474-0.0823i	-0.1550			3	2.1299-0.2139i	-0.3035
		4	1.3865-0.1349i	-0.1801			4	1.4352-0.1466i	-0.1902			4	2.1403-0.3456i	-0.2591
		5	1.3627-0.2952i	-0.2455			5	1.4140-0.3179i	-0.2506			5	2.1778-0.6852i	-0.0972
	2	0	1.3685-0.0874i	-0.2297		2	0	1.4148-0.0974i	-0.2483		2	0	2.1405-0.0631i	-0.2583
		1	1.3159-0.1670i	-0.3709			1	1.3613-0.1849i	-0.3969			1	2.1303-0.0859i	-0.3018
		2	0.2789-1.1387i	-2.0247			2	0.2768-1.1323i	-2.1734			2	2.1181-0.1207i	-0.3537
		3	0.2649-1.4153i	-2.0323			3	0.2629-1.4104i	-2.1809			3	2.1033-0.1771i	-0.4161
		4	0.2547-1.6844i	-2.0376			4	0.2529-1.6807i	-2.1860			4	2.0833-0.2778i	-0.4999
		5	0.2467-1.9488i	-2.0416			5	0.2450-1.9457i	-2.1900			5	2.0365-0.4915i	-0.6927

boundary conditions may lead to qualitatively new spectral features. Moreover, capturing the evanescent modes as well as the QNMs are necessary to describe the evolution of these perturbed systems fully, which necessitates the use of spectral methods. It may be that the extended method could reveal additional branches of solutions. Additionally, explorations with this extended method of more exotic perturbed metrics are also encouraged, such as quantum-inspired black holes and those with multiple horizons. More formally, it would be of interest to understand whether the purely imaginary branch identified here admits an analytic interpretation, for example through an appropriate generalisation of monodromy or CFM techniques.

Code availability

- <https://github.com/dutykh/massive-scalar/>

TABLE IX: *Purely imaginary QNMs for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The corresponding results are obtained through our extended spectral method, utilising 200 polynomials with a precision of 200 digits. In this context, $\Omega = M\omega$ and n represent the dimensionless frequency and the corresponding overtone, respectively, while $\Delta\Omega = \Omega_n - \Omega_{n+1}$.*

μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$
1.30	0	0	0.0000 – 19.86636i	0.25001i	1.40	0	0	N/A	N/A	1.42	0	0	N/A	N/A
		1	0.0000 – 20.11647i	0.25012i			1	N/A	N/A			1	N/A	N/A
		2	0.0000 – 20.36650i	0.25003i			2	N/A	N/A			2	N/A	N/A
		3	0.0000 – 20.61657i	0.25007i			3	N/A	N/A			3	N/A	N/A
		4	0.0000 – 20.86663i	0.25006i			4	N/A	N/A			4	N/A	N/A
		5	0.0000 – 21.11668i	0.25005i			5	N/A	N/A			5	N/A	N/A
		6	0.0000 – 21.36673i	0.25006i			6	N/A	N/A			6	N/A	N/A
		7	0.0000 – 21.61678i	0.25005i			7	N/A	N/A			7	N/A	N/A
	1	0	0.0000 – 20.56989i	0.25033i		1	0	0.0000 – 20.31947i	0.25027i			0	0.0000 – 20.31946i	0.25023i
		1	0.0000 – 20.82023i	0.25034i			1	0.0000 – 20.56981i	0.25033i			1	0.0000 – 20.56979i	0.25033i
		2	0.0000 – 21.07053i	0.25030i			2	0.0000 – 20.82015i	0.25034i			2	0.0000 – 20.82013i	0.25034i
		3	0.0000 – 21.32085i	0.25032i			3	0.0000 – 21.07045i	0.25030i			3	0.0000 – 21.07044i	0.25030i
		4	0.0000 – 21.57115i	0.25031i			4	0.0000 – 21.32077i	0.25032i			4	0.0000 – 21.32076i	0.25032i
		5	0.0000 – 21.82145i	0.25030i			5	0.0000 – 21.57108i	0.25031i			5	0.0000 – 21.57106i	0.25031i
		6	0.0000 – 22.07175i	0.25030i			6	0.0000 – 21.82138i	0.25030i			6	0.0000 – 21.82136i	0.25030i
		7	0.0000 – 22.32204i	0.25029i			7	0.0000 – 22.07168i	0.25030i			7	0.0000 – 22.07166i	0.25030i
	2	0	0.0000 – 20.49205i	0.25055i		2	0	0.0000 – 20.24138i	0.25065i			0	0.0000 – 20.24135i	0.25064i
		1	0.0000 – 20.74281i	0.25076i			1	0.0000 – 20.49194i	0.25056i			1	0.0000 – 20.49191i	0.25056i
		2	0.0000 – 20.99341i	0.25060i			2	0.0000 – 20.74270i	0.25076i			2	0.0000 – 20.74267i	0.25076i
		3	0.0000 – 21.24405i	0.25064i			3	0.0000 – 20.99329i	0.25060i			3	0.0000 – 20.99327i	0.25060i
		4	0.0000 – 21.49468i	0.25064i			4	0.0000 – 21.24394i	0.25064i			4	0.0000 – 21.24392i	0.25065i
		5	0.0000 – 21.74529i	0.25061i			5	0.0000 – 21.49458i	0.25064i			5	0.0000 – 21.49455i	0.25064i
		6	0.0000 – 21.99590i	0.25061i			6	0.0000 – 21.74518i	0.25061i			6	0.0000 – 21.74516i	0.25061i
		7	0.0000 – 22.24650i	0.25060i			7	0.0000 – 21.99580i	0.25061i			7	0.0000 – 21.99578i	0.25061i

Appendix A: Appendix

In Tables XIII and XIV, we explore the case of large values of the rescaled mass parameter; we consider the low multipoles and low overtones for $\mu = 3, 10, 100$. In so doing, we find in Table XIII that propagative and evanescent solutions can co-exist, particularly for the case of $\mu = 10$. When $\mu = 3$, propagative modes arise only for the lowest overtones. For $\mu = 3$ and $\ell = 0$, we detect two QNMs while the evanescent modes exhibit non-monotonic behaviour; for $\ell = 1$, the non-monotonic behaviour of the evanescent modes persists, but differently than before, and we detect one QNM; for $\ell = 2$, we find no QNMs but evanescent modes display a strong non-monotonic behaviour. For $\mu = 10$ and $\ell = 0$, the number of QNMs (eight in total) increases compared to the case of $\mu = 3$ and $\ell = 0$ but evanescent modes still present as non-monotonic. For $\mu = 10$ and $\ell = 1$, we detect eight QNMs in total and the same non-monotonic behaviour for the evanescent modes; for $\ell = 2$, we observe a total of seven QNMs and the non-monotonic evanescent modes. For the $\mu = 100$ case, the extended spectral method struggles to converge to solutions. When $\ell = 0$, we detect only one evanescent mode and no QNMs; we detect neither for $\ell = 1$. For $\ell = 2$, we observe four evanescent modes and no QNMs.

In Table XIV, we once again find purely imaginary modes, with the magnitude of $\Im m[\Omega]$ increasing with n . For $\mu = 3$, we find a fixed spacing between overtones of $\Delta\Omega \sim 0.25004i$ from the onset, whereas for higher multipoles, the

TABLE X: *Evanescent modes (Ω_e) for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\Omega = M\omega$ denotes the dimensionless QNF, n the overtone number, and $\chi^2 = \Re[\omega]^2 - \mu^2$ (with $M = 1$). “N/A” indicates data not available.*

μ	ℓ	n	Ω_e	χ^2	μ	ℓ	n	Ω_e	χ^2	μ	ℓ	n	Ω_e	χ^2
1.45	0	0	1.2835-0.0121i	-0.0426	1.50	0	0	1.3999-0.0162i	-0.0003	2.20	0	0	1.3999-0.0162i	-0.0567
		1	1.2797-0.1748i	-0.0524			1	1.3957-0.0233i	-0.012			1	1.3957-0.0232i	-0.0684
		2	1.2750-0.0263i	-0.0644			2	1.3906-0.0343i	-0.0262			2	1.3906-0.0343i	-0.0826
		3	1.2693-0.0418i	-0.0789			3	1.3849-0.0533i	-0.0421			3	1.3849-0.0533i	-0.0985
		4	1.2637-0.0710i	-0.0931			4	1.3803-0.0885i	-0.0548			4	1.3803-0.0885i	-0.1112
		5	1.2643-0.1357i	-0.0915			5	1.3848-0.1642i	-0.0423			5	1.3848-0.1642i	-0.0987
	1	0	1.2797-0.0138i	-0.0524		1	0	1.3760-0.0180i	-0.0666		1	0	1.3953-0.0189i	-0.0695
		1	1.2742-0.0207i	-0.0664			1	1.3701-0.0266i	-0.0828			1	1.3892-0.0279i	-0.0865
		2	1.2667-0.0326i	-0.0855			2	1.3622-0.0412i	-0.1044			2	1.3812-0.0430i	-0.1087
		3	1.2563-0.0548i	-0.1117			3	1.3517-0.0680i	-0.1329			3	1.3708-0.0707i	-0.1373
		4	1.2415-0.1020i	-0.1487			4	1.3380-0.1235i	-0.1698			4	1.3573-0.1280i	-0.1741
		5	1.2115-0.2305i	-0.2223			5	1.3118-0.2731i	-0.2392			5	1.3321-0.2819i	-0.2419
	2	0	0.3064-0.8742i	-1.5961		2	0	1.2709-0.1497i	-0.3448		2	0	1.2888-0.1566i	-0.3554
		1	0.2854-1.1564i	-1.6085			1	0.3022-0.8584i	-1.8687			1	0.3014-0.8552i	-1.9256
		2	0.2713-1.4287i	-1.6164			2	0.2810-1.1449i	-1.881			2	0.2803-1.1424i	-1.9378
		3	0.2607-1.6948i	-1.6220			3	0.2670-1.4199i	-1.8887			3	0.2662-1.4181i	-1.9455
		4	0.2522-1.9569i	-1.6264			4	0.2567-1.6880i	-1.8941			4	0.2559-1.6866i	-1.9509
		5	0.2450-2.2164i	-1.6300			5	0.2485-1.9516i	-1.8982			5	0.2478-1.9505i	-1.9550

convergence to a fixed spacing occurs as n increases. For higher multipoles, and for the case of $\mu = 10$, we observe that $\Delta\Omega$ decreases with n until it converges to a fixed value. Finally, we note with interest that for $\mu = 100$, the extended spectral method does not capture this imaginary spectrum.

As discussed in Section V, Ref. [8] claims that $\Re[\Omega] \rightarrow \ln[3]/(8\pi)$ as $n \rightarrow \infty$, irrespective of whether the field is light or heavy (where $\mu = 0.3$ and $\mu = 3$ for $\ell = 0$ were considered). Our spectral simulations do not seem to support this behaviour. For $\mu \gtrsim 3$, we instead observe that the real part of the QNFs exhibits a non-monotonic dependence on the overtone index. More precisely, $\Re[\Omega]$ initially decreases, reaches a minimum, and then increases again, with no indication of convergence towards $\ln[3]/(8\pi)$ within the range of overtones that we can reliably resolve.

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TABLE XI: *Purely imaginary QNMs for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The corresponding results are obtained through our extended spectral method, utilising 200 polynomials with a precision of 200 digits. In this context, $\Omega = M\omega$ and n represent the dimensionless frequency and the corresponding overtone, respectively, while $\Delta\Omega = \Omega_n - \Omega_{n+1}$. Here, we find that $\chi^2 = -\mu^2$.*

μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$
1.45	0	0	0.0000 – 19.61622i	0.25001i	1.50	0	0	0.0000 – 19.61618i	0.25000i	2.20	0	0	0.0000 – 19.86540i	0.25008i
		1	0.0000 – 19.86624i	0.25002i			1	0.0000 – 19.86620i	0.25002i			1	0.0000 – 20.11553i	0.25013i
		2	0.0000 – 20.11636i	0.25012i			2	0.0000 – 20.11632i	0.25012i			2	0.0000 – 20.36557i	0.25004i
		3	0.0000 – 20.36639i	0.25003i			3	0.0000 – 20.36635i	0.25003i			3	0.0000 – 20.61566i	0.25009i
		4	0.0000 – 20.61646i	0.25007i			4	0.0000 – 20.61642i	0.25007i			4	0.0000 – 20.86574i	0.25007i
		5	0.0000 – 20.86653i	0.25006i			5	0.0000 – 20.86648i	0.25006i			5	0.0000 – 21.11581i	0.25007i
		6	0.0000 – 21.11657i	0.25005i			6	0.0000 – 21.11653i	0.25005i			6	0.0000 – 21.36588i	0.25008i
		7	0.0000 – 21.36663i	0.25006i			7	0.0000 – 21.36659i	0.25006i			7	0.0000 – 21.61595i	0.25007i
	1	0	0.0000 – 20.31943i	0.25027i		1	0	0.0000 – 20.31938i	0.25026i		1	0	0.0000 – 20.31850i	0.25025i
		1	0.0000 – 20.56977i	0.25034i			1	0.0000 – 20.56972i	0.25034i			1	0.0000 – 20.56887i	0.25038i
		2	0.0000 – 20.82011i	0.25034i			2	0.0000 – 20.82006i	0.25034i			2	0.0000 – 20.81923i	0.25036i
		3	0.0000 – 21.07041i	0.25030i			3	0.0000 – 21.07037i	0.25030i			3	0.0000 – 21.06955i	0.25032i
		4	0.0000 – 21.32073i	0.25032i			4	0.0000 – 21.32069i	0.25032i			4	0.0000 – 21.31989i	0.25034i
		5	0.0000 – 21.57104i	0.25031i			5	0.0000 – 21.57100i	0.25031i			5	0.0000 – 21.57022i	0.25032i
		6	0.0000 – 21.82134i	0.25030i			6	0.0000 – 21.82130i	0.25030i			6	0.0000 – 21.82053i	0.25032i
		7	0.0000 – 22.07164i	0.25030i			7	0.0000 – 22.07160i	0.25030i			7	0.0000 – 22.07085i	0.25032i
	2	0	0.0000 – 20.49187i	0.25056i		2	0	0.0000 – 20.49182i	0.25057i		2	0	N/A	N/A
		1	0.0000 – 20.74263i	0.25076i			1	0.0000 – 20.74257i	0.25076i			1	N/A	N/A
		2	0.0000 – 20.99323i	0.25060i			2	0.0000 – 20.99317i	0.25060i			2	N/A	N/A
		3	0.0000 – 21.24388i	0.25065i			3	0.0000 – 21.24382i	0.25069i			3	N/A	N/A
		4	0.0000 – 21.49452i	0.25064i			4	0.0000 – 21.49446i	0.25064i			4	N/A	N/A
		5	0.0000 – 21.74513i	0.25061i			5	0.0000 – 21.74507i	0.25061i			5	N/A	N/A
		6	0.0000 – 21.99574i	0.25062i			6	0.0000 – 21.99569i	0.25062i			6	N/A	N/A
		7	0.0000 – 22.24634i	0.25060i			7	0.0000 – 22.24629i	0.25060i			7	N/A	N/A

TABLE XII: *QNMs (Ω) for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\Omega = M\omega$ denotes the dimensionless QNF, n the overtone number. QNMs (Ω) corresponding to the parameters of Tables VIII and X exist only for $\ell = n = 0$.*

μ	ℓ	n	Ω	χ^2
1.30	0	0	1.3058-0.3551i	0.0151
1.40			1.4424-0.4156i	0.1205
1.42			1.4424-0.4156i	0.0641
1.45			1.4769-0.4312i	0.0787
1.50			1.5349-0.4576i	0.1059
2.20			2.3906-0.8718i	0.8750

TABLE XIII: QNM s (Ω) and evanescent modes (Ω_e) for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The results are obtained using our extended spectral method with 200 Chebyshev polynomials and 200-digit precision. Here, $\Omega = M\omega$ denotes the dimensionless QNF, and n the overtone number. “N/A” indicates data not available.

μ	ℓ	n	Ω	Ω_e	μ	ℓ	n	Ω	Ω_e	μ	ℓ	n	Ω	Ω_e
3	0	0	$3.0842 - 0.6922i$	$2.9305 - 0.1137i$	10	0	0	$10.0683 - 1.6430i$	$9.8058 - 0.9043i$	10^2	0	0	N/A	$97.9653 - 8.1301i$
		1	$3.4567 - 1.4295i$	$2.9269 - 0.1426i$			1	$10.1768 - 1.8563i$	$9.8249 - 0.9880i$			1	N/A	N/A
		2	N/A	$2.9254 - 0.1820i$			2	$10.3277 - 2.1214i$	$9.8504 - 1.0832i$			2	N/A	N/A
		3	N/A	$2.9287 - 0.2373i$			3	$10.5448 - 2.4628i$	$9.8843 - 1.1924i$			3	N/A	N/A
		4	N/A	$2.9425 - 0.3187i$			4	$10.8737 - 2.9265i$	$9.9291 - 1.3189i$			4	N/A	N/A
		5	N/A	$2.9803 - 0.4489i$			5	$11.4156 - 3.6145i$	$9.9886 - 1.4670i$			5	N/A	N/A
	1	0	$3.1422 - 1.1956i$	$2.9220 - 0.1341i$		1	0	$10.0419 - 1.6272i$	$9.7974 - 0.8961i$		1	0	N/A	N/A
		1	N/A	$2.9177 - 0.1710i$			1	$10.1429 - 1.8380i$	$9.8151 - 0.9790i$			1	N/A	N/A
		2	N/A	$2.9163 - 0.2227i$			2	$10.2829 - 2.0998i$	$9.8387 - 1.0733i$			2	N/A	N/A
		3	N/A	$2.9213 - 0.2985i$			3	$10.4830 - 2.4364i$	$9.8702 - 1.1814i$			3	N/A	N/A
		4	N/A	$2.9408 - 0.4184i$			4	$10.7828 - 2.8923i$	$9.9119 - 1.3066i$			4	N/A	N/A
		5	N/A	$2.9961 - 0.6359i$			5	$11.2669 - 3.5652i$	$9.9675 - 1.4531i$			5	N/A	N/A
	2	0	N/A	$2.9069 - 0.1546i$		2	0	$10.0847 - 1.8111i$	$9.7973 - 0.9651i$		2	0	N/A	$97.9490 - 8.0581i$
		1	N/A	$2.8992 - 0.2010i$			1	$10.2069 - 2.0683i$	$9.8177 - 1.0582i$			1	N/A	$97.9608 - 8.1252i$
		2	N/A	$2.8931 - 0.2684i$			2	$10.3796 - 2.3983i$	$9.8453 - 1.1646i$			2	N/A	$97.9730 - 8.1931i$
		3	N/A	$2.8910 - 0.3730i$			3	$10.6335 - 2.8436i$	$9.8818 - 1.2879i$			3	N/A	$97.9856 - 8.2617i$
		4	N/A	$2.8961 - 0.5539i$			4	$11.0298 - 3.4948i$	$9.9306 - 1.4322i$			4	N/A	N/A
		5	N/A	$2.8910 - 0.9412i$			5	$11.7102 - 4.6002i$	$9.9961 - 1.6036i$			5	N/A	N/A

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TABLE XIV: *Purely imaginary QNMs for massive scalar perturbations of the Schwarzschild black hole for $\ell \in \{0, 1, 2\}$, and different values of the rescaled mass parameter μ . The corresponding results are obtained through our extended spectral method, utilising 200 polynomials with a precision of 200 digits. In this context, $\Omega = M\omega$ and n represent the dimensionless frequency and the corresponding overtone, respectively, while $\Delta\Omega = \Omega_n - \Omega_{n+1}$.*

μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$	μ	ℓ	n	Ω	$\Delta\Omega$
3	0	0	0.0000 – 31.36718i	0.25004i	10	0	0	0.0000 – 22.54378i	0.25222i	10^2	0	0	N/A	N/A
		1	0.0000 – 31.61723i	0.25004i			1	0.0000 – 22.79587i	0.25209i			1	N/A	N/A
		2	0.0000 – 31.86727i	0.25004i			2	0.0000 – 23.04794i	0.25208i			2	N/A	N/A
		3	0.0000 – 32.11731i	0.25004i			3	0.0000 – 23.29990i	0.25196i			3	N/A	N/A
		4	0.0000 – 32.36735i	0.25004i			4	0.0000 – 23.55177i	0.25187i			4	N/A	N/A
		5	0.0000 – 32.61739i	0.25004i			5	0.0000 – 23.80358i	0.25180i			5	N/A	N/A
		6	0.0000 – 32.86743i	0.25004i			6	0.0000 – 24.05530i	0.25173i			6	N/A	N/A
		7	0.0000 – 33.11747i	0.25004i			7	0.0000 – 24.30696i	0.25166i			7	N/A	N/A
	1	0	0.0000 – 20.06658i	0.25058i		1	0	0.0000 – 22.99102i	0.25289i			0	N/A	N/A
		1	0.0000 – 20.31684i	0.25025i			1	0.0000 – 23.24376i	0.25273i			1	N/A	N/A
		2	0.0000 – 20.56729i	0.25046i			2	0.0000 – 23.49632i	0.25257i			2	N/A	N/A
		3	0.0000 – 20.81767i	0.25037i			3	0.0000 – 23.74882i	0.25250i			3	N/A	N/A
		4	0.0000 – 21.06803i	0.25036i			4	0.0000 – 24.00121i	0.25239i			4	N/A	N/A
		5	0.0000 – 21.31841i	0.25038i			5	0.0000 – 24.25350i	0.25229i			5	N/A	N/A
		6	0.0000 – 21.56876i	0.25035i			6	0.0000 – 24.50570i	0.25220i			6	N/A	N/A
		7	0.0000 – 21.81911i	0.25035i			7	0.0000 – 24.75782i	0.25212i			7	N/A	N/A
	2	0	0.0000 – 20.73928i	0.25082i		2	0	0.0000 – 22.90000i	0.25333i			0	N/A	N/A
		1	0.0000 – 20.98992i	0.25064i			1	0.0000 – 23.15334i	0.25334i			1	N/A	N/A
		2	0.0000 – 21.24068i	0.25076i			2	0.0000 – 23.40641i	0.25307i			2	N/A	N/A
		3	0.0000 – 21.49137i	0.25069i			3	0.0000 – 23.65942i	0.25301i			3	N/A	N/A
		4	0.0000 – 21.74205i	0.25068i			4	0.0000 – 23.91236i	0.25294i			4	N/A	N/A
		5	0.0000 – 21.99274i	0.25069i			5	0.0000 – 24.16518i	0.25282i			5	N/A	N/A
		6	0.0000 – 22.24340i	0.25066i			6	0.0000 – 24.41791i	0.25273i			6	N/A	N/A
		7	0.0000 – 22.49405i	0.25065i			7	0.0000 – 24.67056i	0.25265i			7	N/A	N/A

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